Ex. No. 1	**Working with Widgets and Layouts**
Date of Exercise	
Github Code Link (Keep as public)	https://github.com/girishnp17/flutter-lab1-widgets-layouts
Video Link (Google Drive Link)	

Aim

To design and develop a Flutter application that displays a list of fruit items using Row, Column, Container, Card, and ListView widgets, and to create reusable custom widgets to organize the user interface effectively.

Description

Flutter builds its user interface entirely out of widgets that can be composed together. Layout widgets such as Row and Column arrange their children horizontally and vertically, while Container adds padding, margin, decoration and sizing around a child widget. Card gives a widget a material elevation and rounded surface, and ListView efficiently lays out and scrolls a list of children, building only the items that are visible on screen (when used with ListView.builder).

This experiment implements a simple Fruit Shop application. A Fruit data model holds the name, an emoji icon, price per kilogram, a short description and an accent colour for each item. A reusable custom widget, FruitCard, composes a Card containing a Row with a Container used as a circular icon badge, a Column for the name and description (using Expanded so it fills the remaining width), and a rounded Container used as a price chip. The top-level screen, FruitListScreen, places a Container banner above a ListView.builder that renders one FruitCard per fruit in the list, demonstrating how layout widgets and custom reusable widgets combine to build a clean, maintainable UI.

Program

lib/models/fruit.dart

```
import 'package:flutter/material.dart';

class Fruit {
  final String name;
  final String emoji;
  final double pricePerKg;
  final String description;
  final Color color;

  const Fruit({
    required this.name,
    required this.emoji,
```

```
    required this.pricePerKg,
    required this.description,
    required this.color,
  });
}

const List<Fruit> fruits = [
  Fruit(
    name: 'Apple',
    emoji: '🍎',
    pricePerKg: 180,
    description: 'Crisp, sweet and rich in fibre.',
    color: Color(0xFFE57373),
  ),
  Fruit(
    name: 'Banana',
    emoji: '🍌',
    pricePerKg: 60,
    description: 'Soft, energy-rich and easy to carry.',
    color: Color(0xFFFFD54F),
  ),
  Fruit(
    name: 'Grapes',
    emoji: '🍇',
    pricePerKg: 120,
    description: 'Juicy bite-sized fruit, great for snacking.',
    color: Color(0xFF9575CD),
  ),
  Fruit(
    name: 'Orange',
    emoji: '🍊',
    pricePerKg: 90,
    description: 'Tangy citrus fruit loaded with vitamin C.',
    color: Color(0xFFFFB74D),
  ),
  Fruit(
    name: 'Mango',
    emoji: '🥭',
    pricePerKg: 150,
    description: 'The king of fruits — sweet and fragrant.',
    color: Color(0xFFFF8A65),
  ),
  Fruit(
    name: 'Watermelon',
    emoji: '🍉',
    pricePerKg: 40,
    description: 'Refreshing and hydrating summer fruit.',
    color: Color(0xFF81C784),
  ),
  Fruit(
    name: 'Pineapple',
    emoji: '🍍',
    pricePerKg: 70,
    description: 'Tropical fruit with a tangy-sweet flavour.',
    color: Color(0xFFFFF176),
  ),
  Fruit(
    name: 'Strawberry',
    emoji: '🍓',
    pricePerKg: 250,
    description: 'Bright red berry with a sweet aroma.',
    color: Color(0xFFF06292),
  ),
];
```

lib/widgets/fruit_card.dart

```
import 'package:flutter/material.dart';

import '../models/fruit.dart';
```

```
/// Reusable card widget that renders a single [Fruit] using a
/// Card > Row > Column + Container composition.
class FruitCard extends StatelessWidget {
  final Fruit fruit;

  const FruitCard({super.key, required this.fruit});

  @override
  Widget build(BuildContext context) {
    return Card(
      elevation: 3,
      margin: const EdgeInsets.symmetric(horizontal: 12, vertical: 6),
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(14)),
      child: Padding(
        padding: const EdgeInsets.all(12),
        child: Row(
          crossAxisAlignment: CrossAxisAlignment.center,
          children: [
            Container(
              width: 56,
              height: 56,
              alignment: Alignment.center,
              decoration: BoxDecoration(
                color: fruit.color.withValues(alpha: 0.25),
                borderRadius: BorderRadius.circular(12),
              ),
              child: Text(fruit.emoji, style: const TextStyle(fontSize: 30)),
            ),
            const SizedBox(width: 14),
            Expanded(
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  Text(
                    fruit.name,
                    style: const TextStyle(fontSize: 17, fontWeight:
FontWeight.bold),
                  ),
                  const SizedBox(height: 4),
                  Text(
                    fruit.description,
                    style: TextStyle(fontSize: 13, color:
Colors.grey.shade700),
                  ),
                ],
              ),
            ),
            const SizedBox(width: 8),
            Container(
              padding: const EdgeInsets.symmetric(horizontal: 10, vertical: 6),
              decoration: BoxDecoration(
                color: fruit.color,
                borderRadius: BorderRadius.circular(20),
              ),
              child: Text(
                '₹${fruit.pricePerKg.toStringAsFixed(0)}/kg',
                style: const TextStyle(
                  fontSize: 12,
                  fontWeight: FontWeight.bold,
                  color: Colors.white,
                ),
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

lib/screens/fruit_list_screen.dart

```
import 'package:flutter/material.dart';

import '../models/fruit.dart';
import '../widgets/fruit_card.dart';

class FruitListScreen extends StatelessWidget {
  const FruitListScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Fruit Shop'),
        centerTitle: true,
      ),
      body: Column(
        children: [
          Container(
            width: double.infinity,
            padding: const EdgeInsets.all(16),
            color: Theme.of(context).colorScheme.primaryContainer,
            child: Text(
              'Fresh fruits delivered to your door — ${fruits.length} items
available',
              style: const TextStyle(fontSize: 14, fontWeight:
FontWeight.w600),
            ),
          ),
          Expanded(
            child: ListView.builder(
              padding: const EdgeInsets.symmetric(vertical: 8),
              itemCount: fruits.length,
              itemBuilder: (context, index) => FruitCard(fruit: fruits[index]),
            ),
          ),
        ],
      ),
    );
  }
}
```

lib/main.dart

```
import 'package:flutter/material.dart';

import 'screens/fruit_list_screen.dart';

void main() {
  runApp(const FruitShopApp());
}

class FruitShopApp extends StatelessWidget {
  const FruitShopApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Fruit Shop',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.green),
        useMaterial3: true,
      ),
      home: const FruitListScreen(),
    );
  }
}
```

Output Screenshots

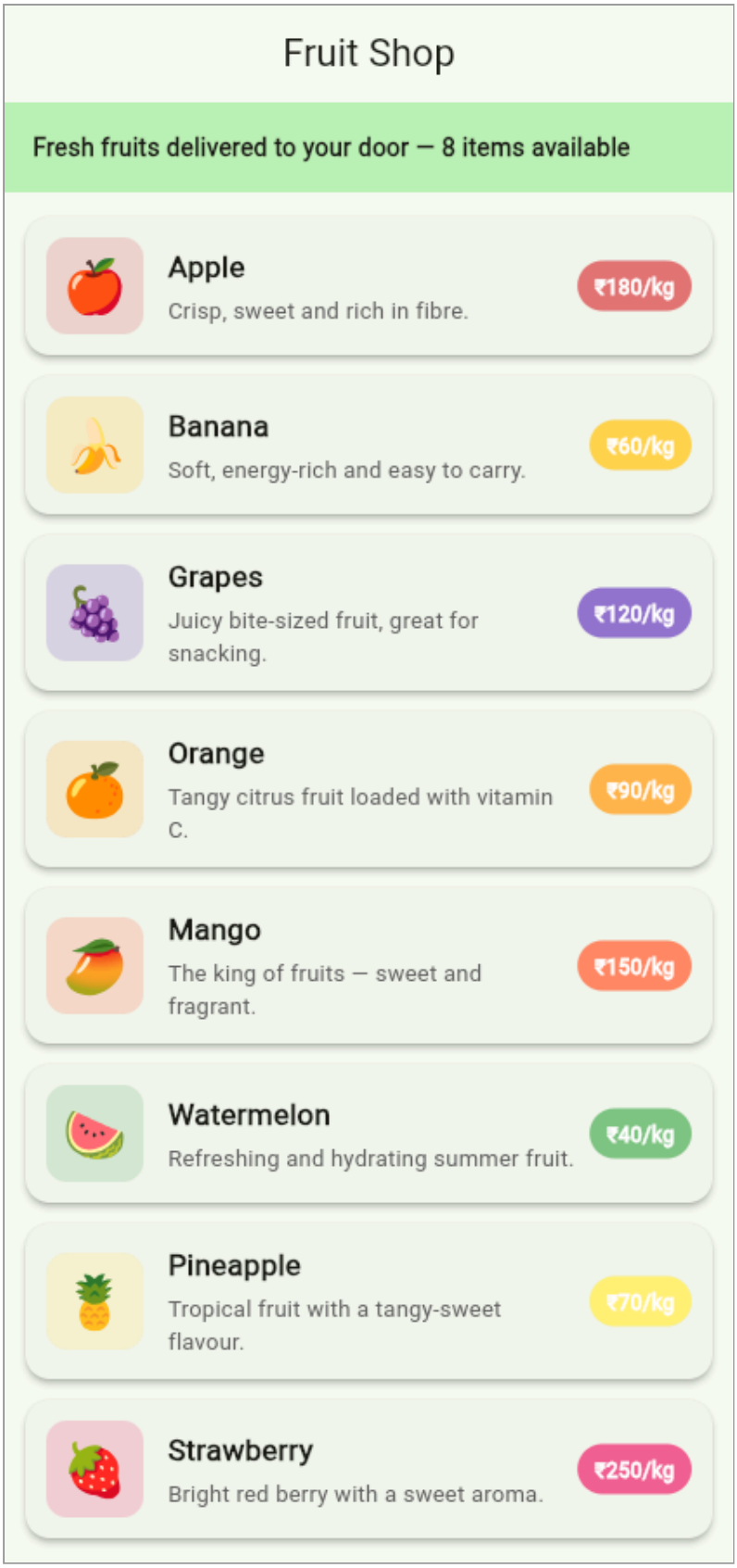


Fig 1. Fruit list rendered with Card, Row, Column and Container widgets inside a ListView (mobile width).

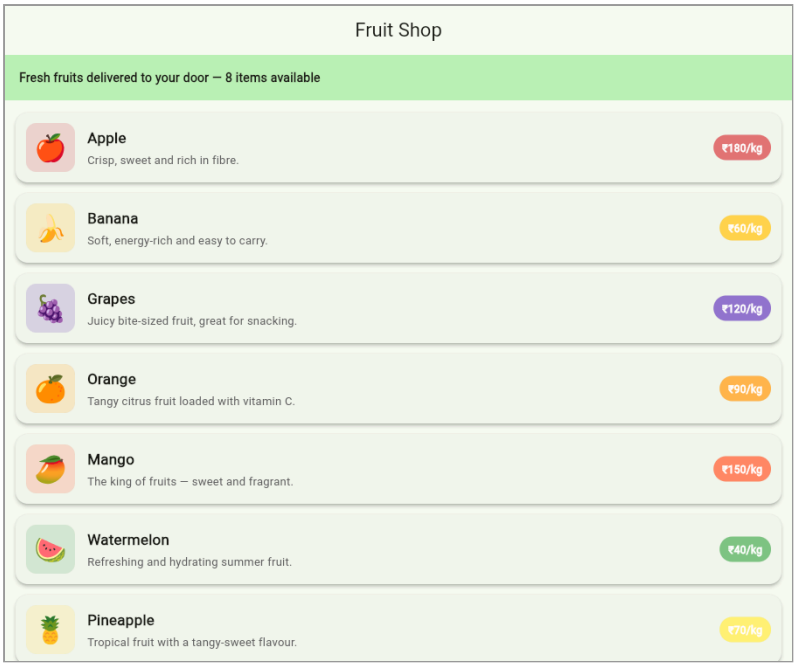


Fig 2. Same ListView of FruitCard widgets rendered at a wider viewport.

Result

The Flutter application was successfully designed and executed. A list of fruit items was displayed using Row, Column, Container, Card and ListView widgets, and a reusable custom widget (FruitCard) was created to organize the UI effectively. The output was verified in the browser as shown in the screenshots above.